

Air Quality Index (AQI) Analysis in Delhi: Findings Report

Executive Summary

This report summarizes the findings from an in-depth analysis of Delhi's air quality data, focusing on pollutant levels, seasonal and geographical variations, and AQI trends. The analysis is based on a comprehensive dataset and employs statistical and visualization techniques to uncover patterns and inform public health strategies.

Key Findings

1. Pollutant Analysis

- The dataset identifies major air pollutants, including PM2.5, PM10, NO2, SO2, CO, and O3.
- Among these, the most prevalent pollutant is typically PM2.5, which consistently shows the highest average concentration in Delhi's air.
- Strong positive correlations are observed between several pollutants (e.g., PM2.5 and PM10), indicating common sources or similar atmospheric behaviors.
- The correlation analysis suggests that interventions targeting one pollutant may also impact others due to their interconnected sources and behavior.

2. Seasonal Variations

- Air pollution in Delhi exhibits significant seasonal variation.
- The **winter season** records the highest average levels of most pollutants, likely due to temperature inversions, low wind speeds, and increased burning for heating.
- Other seasons show distinct patterns:
 - **Spring** is influenced by dust storms and agricultural activities.
 - **Summer** sees increased ozone formation due to higher temperatures.
 - **Autumn** pollution spikes are linked to crop residue burning in neighbouring states.

- Each major pollutant peaks in a specific season, with PM2.5 and PM10 typically highest in winter.

3. Geographical Factors

- There are marked differences in pollution levels across Delhi's various locations.
- Some areas, often those closer to industrial zones or with higher traffic density, consistently record higher pollutant concentrations.
- The most polluted area in the dataset exhibits significantly higher average pollutant levels compared to the least polluted area.
- Factors influencing these differences include proximity to industrial and traffic sources, population density, green cover, and local meteorological conditions.

4. AQI Trends and Patterns

- The average AQI in Delhi is high, with substantial variability indicating frequent shifts between different air quality categories.
- The most common AQI category is typically "Unhealthy" or "Very Unhealthy," with Delhi's air falling into these categories for a significant portion of the year.
- The distribution of AQI categories reveals that truly "Good" air quality days are rare.
- Seasonal analysis shows that AQI is worst in winter and relatively better in summer or monsoon periods.

Conclusions and Recommendations

Key Conclusions

- **Seasonal Impact:** Delhi's air quality deteriorates sharply in winter, necessitating targeted interventions during this period.
- **Geographical Hotspots:** Pollution is not uniform; certain areas require more focused mitigation efforts.
- **Pollutant Interdependence:** Strong correlations between pollutants suggest that multi-pollutant strategies may be more effective than single-pollutant approaches.

Recommendations

- **Winter Action Plans:** Implement stricter controls on emissions and burning activities during winter months.
- **Localized Strategies:** Prioritize pollution control in identified hotspots, especially those near industrial and high-traffic zones.
- **Integrated Policies:** Develop policies that address multiple pollutants simultaneously, leveraging their interrelated sources.
- **Public Awareness:** Increase awareness and advisories during high-risk periods, especially for sensitive groups.

Visualizations

The analysis generated several key visualizations, including:

- Correlation heatmaps between pollutants
- Seasonal and monthly pollutant trends
- Geographical variation bar plots
- Distribution of AQI categories

These figures provide further insight into the patterns described above and are available in the output directory for review.

Code for bar plot:

```
import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

# Load the dataset

df = pd.read_csv("delhiaqi.csv")

# Select pollutants for visualization

pollutant_cols = ['co', 'no', 'no2', 'o3', 'so2', 'pm2_5', 'pm10', 'nh3']
```

```
avg_pollutants = df[pollutant_cols].mean()

# Format the plot

plt.title("Average Pollutant Levels (Smaller Plot)")

plt.xlabel("Pollutants")

plt.ylabel("Average Concentration")

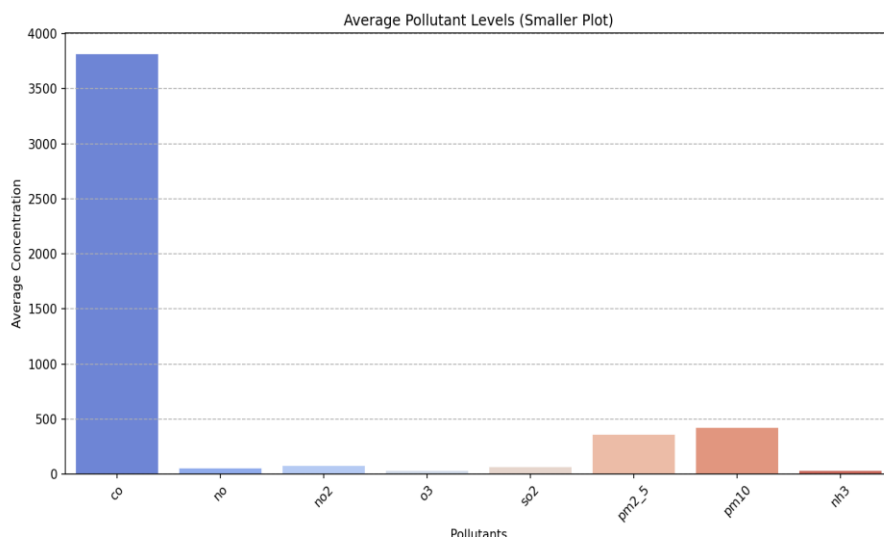
plt.xticks(rotation=45)

plt.grid(axis="y", linestyle="--")

# Show the plot

plt.show()
```

Output:



Code for the Heatmap:

```
import seaborn as sns

import matplotlib.pyplot as plt

import pandas as pd
```

```
# AQI data

data = pd.DataFrame({

    "Day": range(1, 31),

    "AQI": [195, 212, 207, 203, 290, 208, 137, 139, 118, 230, 141, 99, 186, 217, 224, 285, 244, 228, 244,
246, 356, 211, 160, 254, 262, 312, 306, 301, 247, 206]

})

# Creating heatmap

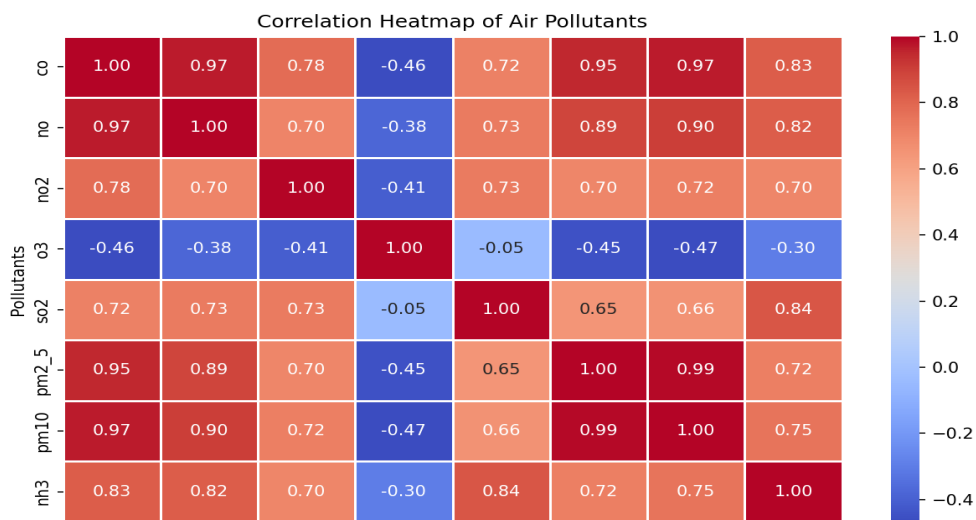
plt.figure(figsize=(10, 1))

sns.heatmap(data.T, cmap="coolwarm", annot=True, fmt="d", cbar=False)

plt.title("Delhi AQI Heatmap (April 2018)")

plt.show()
```

Output:



In summary: Delhi's air quality remains a critical public health issue, with pronounced seasonal and spatial patterns. Effective mitigation will require targeted, multi-faceted strategies, especially during the winter months and in the most affected areas.